Module 10: Foodborne Diseases

Lesson 10.1: Investigation of Foodborne Illness

Lotis M. Balala, DVM, PhD

College of Veterinary Medicine

Visayas State University

Learning Outcomes

Source: Module 10, slide 2

- Define terminologies related to foodborne diseases
- Identify foodborne pathogens and chemical agents that cause food infection and intoxication.
- Discuss the important elements in conducting foodborne illness investigation.
- Identify risk factors predisposing to foodborne illness.
- Describe general and specific signs of foodborne illnesses

Foodborne Illness

Source: Module 10, slide 3

- illness/es acquired through consumption of food and water contaminated with pathogens or certain chemicals
- caused by bacteria, viruses, molds, worms, and protozoa
- chemicals:
  - natural components of foods
  - accidentally added during production and
  - processing
  - carelessness or pollution

The main causes of foodborne illness are:

Source: Module 10, slide 4

- bacterial (66%)
- chemical (26%)
- viral (4%)
- parasitic (4%)

- Food allergy is an abnormal response to a food triggered by the body's immune system.
- Allergens: nuts, milk, eggs, fish, crustacean shellfish, tree nuts, peanuts, wheat or soybeans, can cause allergic reactions in people with food allergies.

https://www.uab.edu/news/youcanuse/item/8792-what-s-making-you-sick-an-in-depth-look-at-food-borne-illnesses

Food allergy illustration

Source: Module 10, slide 5



https://www.nestlenutrition-institute.org/

Classification of foodborne illness

Source: Module 10, slide 6

Foodborne infection

- caused by the ingestion of food contaminated with viable pathogenic organisms that can establish infection
- after ingestion, the organism colonizes the digestive tract and multiplies in number, then spread to other parts of the body via blood circulation

Foodborne intoxication/poisoning

- occurs when pre-formed toxins produced by pathogens are ingested
- may include intoxications from poisonous plants or toxic animal tissues, or due to consumption of food contaminated by chemical poisons

Toxin-mediated infection

- caused by toxin-producing organism but the toxin is produced inside the body (eg. Cl. perfringens)

Food poisoning outbreaks are usually recognized by the occurrence of illness within a short period of time among individuals who have consumed one or more foods in common

Source: Module 10, slide 7

- Diagnosis is suggested by epidemiological findings
- Single cases of food poisonings are difficult to identify unless, as in botulism, there is a distinctive clinical syndrome

Food infection

Source: Module 10, slide 8

- Campylobacter jejuni

Escherichia coli 0157:H7

- Hepatitis A virus
- Listeria monocytogenes
- Norwalk virus
- Salmonella
- Shigella
- Vibrio parahaemolyticus and other marine Vibrio spp.

Food intoxication

- Bacillus cereus
- Clostridium botulinum
- Clostridium perfringens (cafeteria germ)
- Staphylococcus aureus
- toxigenic molds (mycotoxins), poisonous mushrooms, and biogenic amines

Investigation of Foodborne Illness

Source: Module 10, slide 9

Aims of an investigation of foodborne illness

- identify offending foods and beverages
- elucidate information about causative agents and their source
- determine the factors that contribute to growth and survival of causative agents

The investigation involves

Source: Module 10, slide 10

Patient

Food

Place where the food is processed, prepared and served

Patient

Source: Module 10, slide 11

- record the clinical symptoms of the
- affected persons
- determine the incubation period
- obtain clinical specimens as soon as possible such as vomitus, stool and/or blood according to signs and symptoms

Food

Source: Module 10, slide 12

- collect information about all meals and snacks eaten 72 hour before onset of illness
- collect leftovers from food that were eaten 72 hour before onset of illness or some ingredients that were used in foods and of any food available from an allegedly contaminated lot

Place

Source: Module 10, slide 13

- Investigate the hygienic condition at the place where the suspect foods were processed, prepared and served.
- Try to identify the circumstances that permitted survival and growth of food-borne pathogens such as:
- i) inadequate refrigeration
- ii) inadequate cooking and reheating
- iii) holding foods at warm temperature several hours before serving
- iv) inadequate cleaning and disinfection of equipment
- v) the possibility of cross contamination from raw to cooked foods
- Arrange for clinical and laboratory examination of workers who ate the suspect food.
- Submit a report covering all information obtained.

Summary of steps in the investigation of food-borne disease outbreak investigation

Source: Module 10, slide 14

- Verify the existence of an outbreak
  - Compare the current number of cases with the past
  - Note: -consider seasonal variations
- 2. Verify the diagnosis
  - Review clinical and laboratory findings
- 3. Describe the outbreak with respect to time, place and person
- 4. Prepare an epidemic curve
- 5. Calculate food-specific attack rates

- 6. Formulate and test hypotheses (by consulting with higher level health professionals).

7. Search for additional cases

Source: Module 10, slide 15

- 8. Analyze the data
- 9. Make a decision on the hypotheses tested (by consulting with higher level health professionals).
- 10. Intervene and follow-up
- 11. Report the investigation
- 12. Inform the public on the control and prevention of the outbreak

Top 5 Foodborne Illness Risk Factors

Source: Module 10, slide 16

1. Poor Personal Hygiene

- Poor personal hygiene practices serve as the leading cause of foodborne illnesses. Food establishments must promote a culture of food safety by developing an employee illness policy, proper handwashing procedure, and a no barehand contact policy with ready to eat foods

2. Improper Holding Temperatures

- Cold foods shall maintain ≤41°F and hot foods shall maintain ≥135°F. If foods stay in between those temperatures for any period, ensure they are time and temperature controlled and documented. Time and temperature control for safety (TCS) foods must be date marked and stored no longer than 7 days. Remember, first day of preparation or when a food package is opened, counts as day 1. When in doubt, throw it out!

3. Improper Cooking Temperatures

Source: Module 10, slide 17

- Ensure metal stem thermometers are conveniently stored and accessible for employees to monitor final cooking temperatures.

4. Food from Unsafe Sources

- All foods distributed in licensed food establishments must be obtained from approved sources that comply with applicable laws and regulations. All food shall be inspected upon delivery to ensure proper temperatures, it’s condition and overall sanitation.

5. Contaminated Equipment/Cross-Contamination

Source: Module 10, slide 18

- Cross contamination can occur from a variety of sources, including chemical and raw foods.
- To prevent the spread of harmful pathogens
  - all equipment and utensils shall be properly cleaned and sanitized at least once every 4 hours
  - ensure sanitizer solutions are monitored using their appropriate test strips
  - low sanitizer concentrations will fail in effectively removing pathogens from surfaces, while high concentrations will leave a toxic residue

Effects of food poisoning

Source: Module 10, slide 19

- develop symptoms after ingesting a pathogen
- others never know that they are suffering from food-borne illness
- High risk:
  - compromised immune systems
  - elderly
  - very young

Symptoms of foodborne illnesses depend on the cause

Source: Module 10, slide 20

- Common symptoms of many foodborne illnesses include vomiting, diarrhea or bloody diarrhea, abdominal pain, fever, and chills
- Foodborne illnesses may lead to dehydration, hemolytic uremic syndrome (HUS), and other complications
- Acute foodborne illnesses may also lead to chronic- or long lasting-health problems
- Treatment: replacing lost fluids and electrolytes to prevent dehydration.
- Foodborne illnesses can be prevented by properly storing, cooking, cleaning, and handling foods.

Module 10: Foodborne Diseases

Lesson 10.2: Bacterial, Viral, Parasitic, and Chemical Foodborne Diseases

Lotis M. Balala, DVM, PhD

College of Veterinary Medicine

Visayas State University

Learning Outcomes

Source: Module 10, slide 22

- Discuss etiologic agent, transmission, implicated foods and means of control for specific foodborne diseases.
- Tabulate foodborne diseases based on classification.

Why do foodborne diseases emerge?

Source: Module 10, slide 23

- increase in international travel and trade
- microbial adaptation
- changes in the food production system
- changes in human demographics and behavior

Other contributing factors

Source: Module 10, slide 24

The globalization of the food supply

- Importation of contaminated raspberries from South America led to a large outbreak of cyclosporiasis in North America in 1996-97

Inadvertent introduction of pathogens into new geographic areas

- Introduction of Vibrio cholerae into waters off the coast of southern United States when a cargo ship discharged contaminated ballast water in 1991
- Travelers, refugees, and immigrants exposed to unfamiliar foodborne hazards while abroad
  - International travelers may become infected by foodborne pathogens that are uncommon in their countries
  - 90% of all cases of salmonellosis in Sweden are imported

Changes in microorganisms

Source: Module 10, slide 25

- evolution of new pathogens
- development of new virulent strains in old pathogens
- development of antibiotic resistance pathogens
- ability to survive in adverse environmental conditions

Change in the human population

- population of highly susceptible persons is expanding world-wide
- ageing, malnutrition, HIV infections and other underlying medical conditions
- reduced immunity due to poor nutritional status render people more susceptible to foodborne infections

Changes in lifestyle

Source: Module 10, slide 26

- Greater numbers of people go out and eat meals prepared in restaurants, canteens, fast food outlets, and by street food vendors
- Boom in food service establishments is not matched by effective food safety education and control
- Unhygienic preparation of food provides ample opportunities for contamination, growth, or survival of foodborne pathogens

Foodborne disease source material - slide 27

DISEASE	PATHOGEN	INCUBATION PERIOD/MEAN	DURATION OF ILLNESS	SYMPTOMS	FOODS IMPLICATED	TYPE OF ILLNESS	RESERVOIR
Salmonellosis	*Salmonella* (facultative) BACTERIA	8-72 hours 18-36 hours	2-3 days	Abdominal pain, headache, fever, nausea, diarrhea, chills, cramps	Poultry, eggs, meat, meat products, milk, smoked fish, protein foods, juice	Infection	Domestic and wild animals, humans
Shigellosis	*Shigella* (facultative) BACTERIA	1-7 days 1-3 days	Indefinite	Diarrhea, fever, chills, cramps, lassitude, nausea, dehydration	Moist, mixed foods, milk, potato, tuna, bean, and macaroni salads, apple cider	Infection	Human feces, flies
Listeriosis	*Listeria monocytogenes* BACTERIA	2 days – 3 weeks 4-21 days	Indefinite, high fatality in the immunocompromised	Nausea, vomiting, chills, headache, fever, backache, meningitis	Unpasteurized dairy foods, vegetables, meat, poultry, seafood, raw and smoked fish, ready-to-eat foods	Infection	Domestic and wild animals, humans, soil
Staphylococcus	*Staphylococcus aureus* BACTERIA	1-6 hours 2-4 hours	1-2 days	Nausea, cramps, vomiting, dehydration	Reheated foods, ham, poultry, dairy foods, custards, potato salad, cream-filled foods	Intoxication	Humans (skin, throat, nose), animals
Clostridium perfringens	*Clostridium perfringens* BACTERIA	8-22 hours 10 hours	24 hours	Abdominal pain, diarrhea, gas	Meat, meat products, gravy, thick foods that are held at a low temp or cooled slowly	Infection/ Intoxication	Human intestinal tract, animals, soil
Bacillus cereus	*Bacillus cereus* BACTERIA	½ - 5 hours; 8-16 hours ½ -5 hrs; 12 hrs	6-24 hours; 12 hours	Nausea and vomiting; Diarrhea, abdominal cramps	Rice, custards, seasonings, dry food mixes, spices, salads, casseroles, milk, sauces,	Intoxication	Soil, dust

Foodborne disease source material - slide 28

					meats		
Campylobacter	*Campylobacter jejuni* BACTERIA	2-5 days	5-7 days 3-5 days	Diarrhea which may be watery or bloody, fever, nausea, headache	Chicken, unpasteurized milk; water	Infection	Domestic and wild animals (intestinal tract)
E. coli O157:H7	*Escherichia coli* BACTERIA	2-5 days	5-10 days 2-5 days	Diarrhea (may be bloody), abdominal pain, nausea, vomiting, fever, HUS	Undercooked ground beef, imported cheeses, unpasteurized milk or juice, cider, alfalfa sprouts	Infection/ Intoxication	Human intestinal tract, animals (especially cattle)
Botulism	*Clostridium botulinum* BACTERIA	12-36 hours; 72 or more hours	Several days – year 18-36 hours	Vertigo, inability to swallow, respiratory paralysis, visual disturbances	Improperly processed low-acid canned goods, garlic in oil, grilled onions, meat, stew, baked potatoes	Intoxication	Soil, water
Viral Gastroenteritis	Snow Mountain, calicivirus virus	24-48 hours	24-60 hours 36 hours	Diarrhea, vomiting, nausea, cramps, headache, muscle aches, tiredness, slight fever	Water, ready-to-eat foods (salad, sandwiches, bread) shellfish	N/A	Human intestinal tract
Hepatitis A	Hepatitis A VIRUS	7-50 days	Several weeks to months 25-30 days	Fever, fatigue, stomach pain, nausea, dark urine, jaundice	Ready-to-eat foods, fruit and juice, milk products, shellfish, salads, vegetables, sandwiches, water	N/A	Infected people
Norovirus infection	Noro Virus	12-96 hours	1-2 days	Low-grade fever, chills, headache, muscle ache, and fatigue	Ready-to-eat foods (salad, sandwiches, bread) shellfish	Infection	Human intestinal tract

Also read PDF copy of:

Source: Module 10, slide 29

Also read PDF copy of:

Foodborne Illness Quick Reference

Page 1	General Information
Page 2	Anisakiasis
Page 3	Bacillus cereus
Page 4	Botulism
Page 5	Campylobacteriosis
Page 6	Ciguatera Fish Poisoning
Page 7	Clostridium Perfringens
Page 8	E. coli O157:H7
Page 9	Food Allergens
Page 10	Hepatitis A

Lesson 9.2: Microbial Spoilage of Canned Food

Microbial Spoilage of Canned Food

Source: Lesson 9.2 Microbial Spoilage of Canned Food.pdf, PDF page 1

Lesson 9.2

Microbial Spoilage of Canned Food

Learning outcomes

Source: Lesson 9.2 Microbial Spoilage of Canned Food.pdf, PDF page 2

1.Discuss mechanism of microbial spoilage of canned foods.

2.Illustrate physical signs of spoiled canned foods.

Factors Affecting Microbial Spoilage

Source: Lesson 9.2 Microbial Spoilage of Canned Food.pdf, PDF page 3

- Spoilage of canned goods may be due to

a. Microbes

b. Chemicals

c. Deleterious influences such as rust and damage

- Canned goods are classified as spoiled when the food has undergone a deleterious change or when the condition of the container renders such change possible.
- Spoiled cans may usually show obvious abnormalities such as distortion, blowing, concave ends or slightly constricted sides; or they may present a perfectly normal external appearance.

Microbial spoilage

Source: Lesson 9.2 Microbial Spoilage of Canned Food.pdf, PDF page 4

Type of microorganisms

- bacteria of decomposing or fermenting type are the most important cause while the spore forming bacteria are the most resistant
- three main types of spore forming organism, which can resist normal processing and may cause spoilage in canned foods:

a. Gas producing anaerobic organisms growing at an optimum temperature of 55°C .

b. Gas producing anaerobic and aerobic organisms with an optimum temperature growth of 37°C.

Both of the above types are associated with the form of spoilage recognized by blowing or swell.

c. Non-gas producing aerobic or facultative anaerobic spore- forming organisms with an optimum growth temperature of about 55°C, which produce ‘flat sours’.

Factors determining microbial unsoundness

Source: Lesson 9.2 Microbial Spoilage of Canned Food.pdf, PDF page 5

The factors determining microbial, unsoundness in canned goods are

The number of microorganisms present

- Processing is not a substitute for cleanliness; a small number of bacteria are likely to be destroyed more easily by processing than are a large numbers.

Efficiency of processing

- Insufficient processing must be regarded as a cause of unsoundness of canned foods

Access of the air to the can

- The bacteria found in the canned meat are nearly always secondary invaders gaining access to the food through a leaking tin.
- Microbial spoilage may thus results from under processing or leakage through the tin.

Flat Souring

Source: Lesson 9.2 Microbial Spoilage of Canned Food.pdf, PDF page 6

- manifested by the presence of a sour odor of foodstuffs, but without the can becoming blown.
- Canned foods susceptible to this deleterious change are those containing sugar or starches, and meat products such as sausages or paste containing cereals.
- True flat sours are commonly caused by the growth of thermophilic organisms (B. coagulans, B. stearothermophilus, B.

circulans), which attack carbohydrates with production of acid but not gas.

- The exceptional heat resistance of such organisms makes them to survive the normal processing given to the canned foods.

Flat Souring (continued)

Source: Lesson 9.2 Microbial Spoilage of Canned Food.pdf, PDF page 7

- Sourness in canned foods may also arise due to leaking cans or it may have developed in the food before processing.

- In corned beef, the presence of souring without gas production may be due to under-processing or leaking seams, but the development of pre-processing storage is also likely to be the cause.
- flat sours of canned foods due to thermophilic spore forming organisms can not be detected until the can is opened and its content examined
- unlikely to occur in temperate climates but common in tropical and sub-tropical countries or when cans imported to temperate climate from hot climates

Chemical spoilage - Hydrogen swell

Source: Lesson 9.2 Microbial Spoilage of Canned Food.pdf, PDF page 8

- most important problems encountered in the canning industry owing to spoilage on account of chemical changes such as:
- Hydrogen swelling
- Purple staining
- Rust or damage

Hydrogen Swell

- happens independent of microbial spoilage, and is associated with the formation of hydrogen gas in the can following internal corrosion
- imperfection or scratches on the inner coating may expose small areas of steel and when the contents are acid, an electric coupling may result, the reaction is producing hydrogen gas

Hydrogen swell (continued)

Source: Lesson 9.2 Microbial Spoilage of Canned Food.pdf, PDF page 9

- Can affected with hydrogen swell may show varying degrees of bulging from flipping to blowing and if the tin being punctured, there is emission of hydrogen gas, whichis odorless and burns on the application of a flame
- This condition is chiefly associated with food containing organic acids particularly fruits such as plums, cherries, raspberries, etc.
- The range of acidity most favorable to the production of hydrogen swells lies between pH 3.5 and 4.5.
- This condition is unknown in canned meat foods, but it is sometimes seen in sardines.

Judgement

Source: Lesson 9.2 Microbial Spoilage of Canned Food.pdf, PDF page 10

- Though the content of the can in hydrogen swell may be quite harmless, the routine methods employed in the examination of canned goods render it impossible to distinguish between the tins which are blown due to hydrogen swell and those which are blown as a result of deleterious changes due to bacteria or yeasts.
- So, all blown tins, leakers, springers and flat-sours together with tins whose contents show evidence of mold must be regarded as unfit for food, hence, should be condemned.

Sulphiding

Source: Lesson 9.2 Microbial Spoilage of Canned Food.pdf, PDF page 11

Purple staining on the inner surface of the cans in which sulphur- containing foods are packed may occur with all fish and meat products, especially, liver, kidneys and tongue.

- due to the breaking down of sulphur containing proteins at a high processing temperature with a result that, hydrogen sulphide is liberated and a thin layer of tin sulphide is formed on the inside of the can

This discoloration, which does not involves foodstuff itself and varies from a light pink to a dark purple but a blackening of both inside of the can and the surface of the foodstuffs may accompany it.

These latter changes are due to attack of hydrogen sulphide on the steel base with a formation of iron sulphide.

It is of more serious impart than the deposition of tin sulphide as it may lead to pitting of the steel and disfigurement of the surface of the meat pack.

Prevention, Treatment and Judgement

Source: Lesson 9.2 Microbial Spoilage of Canned Food.pdf, PDF page 12

- Prevention or Treatment of the Can

The basis of which is, copal gum, dissolved in a suitable solvent to which an added substance capable of uniting with volatile “sulphur” gases released while the food is being processed.

Judgement

Purple staining of the inside of the can is of no significance, while blackening of surface of the foodstuffs may be removed by trimming.

Rust or Damage

Source: Lesson 9.2 Microbial Spoilage of Canned Food.pdf, PDF page 13

• external rust requires careful consideration and its conditions beneath can labels should be checked, esp.

that adhesive contains hygroscopic substances

Judgement

• Can in which the external surface rusted without noticeable pitting of iron may be released for immediate sale and consumption.

• But if the rust is removed with a knife and inspection with a hand lens reveals the iron plate to be definitely pitted, there is dangerous of early perforation and the can should be condemned.

• Minute perforations of the plate to spoilage of the can content.

Pin holing and self-sealing cans

Source: Lesson 9.2 Microbial Spoilage of Canned Food.pdf, PDF page 14

• Pin holing may originate from the outside but also from the inside of the can where the tin plating is imperfect or has been fractured during seaming, and in this case lacquer lining aggravates the trouble, as the cracks which occur in the lacquer aids in concentrating the chemical action on a small area.

• A can, which is a leaker or pin holed, may occasionally seal itself by blocking of holes with the contained foodstuffs, an may then proceed to blow up.

• Such self-sealing can may blow early in its life, generally within the first few months.

Common can distortions

Source: Lesson 9.2 Microbial Spoilage of Canned Food.pdf, PDF page 15

The various distortions commonly encountered are

• Swell or blower

•A can with it ends bulged by positive internal pressure due to gas generated by microbial or chemical activity is termed as swell or blower.

• Flipper

•A flipper is a can of normal appearance in which one end flips out when the can is struck against a solid object but the end snaps back to the normal when very slight pressure is applied.

Unfilled cans and damage

Source: Lesson 9.2 Microbial Spoilage of Canned Food.pdf, PDF page 16

• Unfilled cans

• Where unfilled cans are stored and allowed to rust internally before being filled, the can edges may become rusted, with the result that during processing a chemical action may take place between the rust and meat juices and give rise to an unsightly grey precipitate of iron phosphate in the meat jelly.

• Damage

• Considerable significance should be attached to cans damaged by rough handling, the important factor in their judgement being, the extent and location of the damage.

• Marked deformation of the seam of the can is attended by considerable risk of leakage.

Judgement of damaged cans

Source: Lesson 9.2 Microbial Spoilage of Canned Food.pdf, PDF page 17

Judgement

• Can showing seam damage should be condemned.

• Slight indentations on the can body are permissible, but severe dents on the body may cause seam distention with danger of leakage and such can should be rejected.

• Any can having a dent at one end should also be rejected for it is possible to reduce a springer to normal, at any rate temporarily, by hitting it upon the corner of a box.

• Nail holes in cans cause during closing of packaging cases may also be encountered, and such cans should be rejected even if the contained foodstuffs appear perfectly normal.

• It is a wise action to reject any can which is in the least suspicious or which shows lack of concavity of the end.

Springer and Leaker

Source: Lesson 9.2 Microbial Spoilage of Canned Food.pdf, PDF page 18

• Springer

A springer is the term used to describe a can in which one end is bulged but can be forced back into normal position, where upon the opposite end bulges.

• Leaker

A leaker is a can containing a perforation from any cause whereby atmospheric air may enter the can or its contents escape.

May be detected by disappearance of vacuum from the sides and ends of the cans and the bubbles that appear

from the can when held under water and squeezed.

The commonest leaking is at the seams, and may sometimes be detected by the appearance of liquid or stain on the can surface.

Mold formation on the surface of canned meat is also indicative of leakage through the seams but cannot be detected until the can is opened.

Examination of containers

Source: Lesson 9.2 Microbial Spoilage of Canned Food.pdf, PDF page 19

- examination of the quality of containers is important to ensure that there are no damaged, rusty, blown, etc. cans
- for non-shelf-stable products, five containers are examined visually and their contents examined microbiologically
- Both aerobic and anaerobic techniques are undertaken, decisions as to rejection or approval being based on bacterial plate counts (Sampling and Inspection Procedures for Microbiological Examination of Meat Products in Hermetically Sealed Containers, Codex Alimentarius Commission of FAO/WHO 1983).

Foodborne Illness: Disease Reference

New Mexico Environment Department

Source: foodborne_illness_ref.pdf, PDF page 1

Foodborne Illness Quick Reference

Page 1 General Information

Page 2 Anisakiasis

Page 3 Bacillus cereus

Page 4 Botulism

Page 5 Campylobacteriosis

Page 6 Ciguatera Fish Poisoning

Page 7 Clostridium Perfringens

Page 8 E. coli O157: H7

Page 9 Food Allergens

Page 10 Hepatitis A

Page 11 Listeriosis

Page 12 Norovirus

Page 13 Salmonellosis

Page 14 Scombroid Poisoning

Page 15 Shigellosis

Page 16 Staphylococcus

Page 17 Vibrio spp.

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General Information

Source: foodborne_illness_ref.pdf, PDF page 2

Classification of FBI's

- Foodborne Infection-
  - After ingestion the organism burrows into the digestive tract and multiplies in number.
  - Can spread to other parts of the body through the blood stream.
  - Bacteria, viruses, and parasites can cause FB infections.
  - Example: Salmonellosis.
- Intoxication-
  - Caused when a living organism multiplies in or on a food and then produces a chemical waste or toxin.
  - After consumption the toxin causes illness (typically called food poisoning).
  - An intoxication may also be caused by man-made chemicals (bleach)
  - Examples: Clostridium botulinum and Staphylococcus aureus.
- Toxin-mediated infection-
  - Caused by consuming a living organism, which then produces a toxin that causes illness.
  - Differs from an intoxication because the toxin is produces inside the body.
  - Example: Clostridium perfringens.

Bacteria

- All bacteria exist in a vegetative form
  - Vegetative cells grow, reproduce, and produce wastes just like other living organisms.
- Some bacteria have the ability to form spores
  - Spores help bacteria survive when the environment is too hot, cold, dry, acidic, or when there is not enough food.
  - Spores are not able to grow reproduce.

Vegetative Cells (Optimal Conditions)

Spores (Stress Conditions)

Yes

Yes

Yes

No

Yes

No

No

No

Yes

No

Reproduce

Grow

Produce Toxin

Resistant to Stress

Harmful if Eaten

Temperature Conditions

Source: foodborne_illness_ref.pdf, PDF page 3

- Psychrophilic bacteria grow within the range of 32°F to 70°F
  - Most are spoilage organisms, but some cause illness
- Mesophilic bacteria grow within the range of 70°F to 110°F
- Thermophilic bacteria grow above 110°F
  - All thermophilic bacteria are spoilage organisms

Oxygen

- Aerobic bacteria must have oxygen to grow.
- Anaerobic bacteria can not grow when oxygen is present.
  - Anaerobic bacteria grow well in vacuum packaged foods.
  - Anaerobic bacteria also grow well in the middle of cooked food masses.
- Facultative anaerobic bacteria can grow with or without free oxygen.
  - Most foodborne illness causing microorganisms are facultative anaerobic.
- Microaerophilic organisms have very specific oxygen requirements, usually around 3-6 percent.

Bacteria

Bacillus cereus

Campylobacter jejuni

Clostridium perfringens

Clostridium botulinum

Escherichia coli

Listeria monocytogenes

Salmonella spp.

Shigella spp.

Staphylococcus aureus

Vibrio spp.

Viruses

Hepatitis A

Norwalk virus group

Rotavirus

Parasites

Anisakis spp.

Cryptosporidium parvum

Cyclospora cayetonensis

Giardia lamblia

Toxoplasma gondii

Trichinella spiralis

Anisakiasis

Source: foodborne_illness_ref.pdf, PDF page 4

Pathogen: Anisakis spp.

Type: Parasitic infection

Spore Former: No

Description: Anisakis spp. Are nematodes (roundworms) found in fish. They are usually 1 to 1 ½ inches long and the diameter of a human hair. They are beige, ivory, white, gray brown, or pink.

Incubation Period: 1 hour to 2 weeks

Duration of Illness: less than 3 weeks

Symptoms:

-If the worm attaches to the throat: Coughing.

-If the worm attaches to the stomach: vomiting and abdominal pain.

-If the worm attaches to the large intestine: Sharp pain and fever, similar to appendicitis.

Common Cause: The worms are transferred to fish (mostly bottom feeders) through seawater.

Foods Implicated: Raw, undercooked, or improperly frozen fish, including cod, haddock, fluke, salmon, herring, flounder, monkfish. Sea urchins, crab, shrimp, starfish, and tuna.

Prevention: Purchase seafood from approved dealers. Cook seafood to proper temperatures. Properly freeze fish.

Bacillus cereus

Source: foodborne_illness_ref.pdf, PDF page 5

Pathogen: Bacillus cereus

Type: Bacterial intoxication or Toxin-mediated infection

Oxygen: Facultative anaerobic

Spore Former: Yes

Incubation Period: Vomit type: 30 minutes to 6 hours. Diarrhea type: 6 to 16 hours.

Duration of Illness: Both less than 24 hours, 12-14.

Symptoms: Vomit type: Nausea and vomiting, occasionally abdominal cramps and/or diarrhea. Diarrhea type: Watery diarrhea, abdominal cramps and pain, nausea.

Common Cause: Most often associated with foods that are improperly stored (in the “Danger Zone”), which allows spores to convert into vegetative cells. Vegetative cells then produce toxin which leads to illness.

Foods Implicated: Vomit type: usually associated with grain products such as rice, potatoes, pasta, corn, cornstarch, soybeans, tofu, and flour. Diarrhea type: meats, milk, vegetables, and fish.

Prevention: Foods must be cooked and hot held properly and then cooled down quickly within 4 hours.

Botulism

Source: foodborne_illness_ref.pdf, PDF page 6

Pathogen: Clostridium botulinum

Type: Bacterial intoxication

Oxygen: Anaerobic

Spore Former: Yes

Incubation Period: 12 to 36 hours, but may vary from 4 hours to 8 days

Duration of Illness: Several days to a year

Symptoms: Dizziness, blurred or double vision, difficulty speaking, swallowing, and breathing, fatigue, weakness, headache.

Common Cause: Improperly heat-processed foods, especially home-canned products.

Foods Implicated: Low-acid foods that are inadequately heat-processed, packaged anaerobically, and held in the “Danger Zone”.

Prevention: Do not use home-canned foods or foods from unapproved sources and discard bulging cans.

Campylobacteriosis

Source: foodborne_illness_ref.pdf, PDF page 7

Pathogen: Campylobacter jejuni

Type: Bacterial infection

Oxygen: Microaerophilic (3-6% oxygen)

Spore Former: No

Incubation Period: 2 to 5 days

Duration of Illness: 2 to 7 days, usually no more than 10 days (relapses common)

Symptoms: Diarrhea (watery and bloody), abdominal pain, fever, nausea and vomiting, headache, muscle pain.

Common Cause: Often transferred from raw meats to other foods by cross contamination, often via food contact surfaces such as a cutting board or food worker’s hands.

Foods Implicated: raw milk, raw poultry, raw meats, non-chlorinated water.

Prevention: Thoroughly cook food to safe minimum internal temperatures, avoid cross contamination, clean and sanitize food contact surfaces.

Ciguatera Fish Poisoning

Source: foodborne_illness_ref.pdf, PDF page 8

Pathogen: Ciguatoxin

Type: Biochemical intoxication

Oxygen:

Spore Former: No

Incubation Period: 15 minutes to 1 day

Duration of Illness: A few days, but may take several weeks or months

Symptoms: Dizziness, hot and cold flashes, diarrhea, vomiting, headache.

Common Cause: The toxin is transferred to finfish when they eat algae or other fish that contain the toxin.

Foods Implicated: Fish, including barracuda, grouper, snapper, jack, mackerel, hogfish, amberjack, blackjack, and triggerfish.

Prevention: Purchase seafood from approved dealers. Cooking will not destroy toxin.

Clostridium perfringens

Source: foodborne_illness_ref.pdf, PDF page 9

Pathogen: Clostridium perfringens

Type: Toxin-mediated infection

Oxygen: Anaerobic

Spore Former: Yes

Incubation Period: 8 to 22 hours

Duration of Illness: 24 hours

Symptoms: Abdominal pain, diarrhea, nausea, dehydration, (fever and vomiting usually absent).

Common Cause: Most often attributed to foods that are temperature abused, especially those that have been improperly cooled and reheated.

Foods Implicated: Cooked meats and poultry that have been improperly cooled and served cold or improperly reheated, stews, gravy, beans that have been cooled slowly.

Prevention: Foods must be cooled down quickly within 4 hours, reheated quickly to 165°F within 2 hours, and then held above 140°F.

E. coli O157:H7 enteritis

Source: foodborne_illness_ref.pdf, PDF page 10

Pathogen: Escherichia coli O157:H7

Type: Bacterial infection or toxin-mediated infection

Oxygen: Facultative anaerobic

Spore Former: No

Incubation Period: 12 to 72 hours

Duration of Illness: 1 to 3 days

Symptoms: Serious threat for children up to age 16 and the elderly. Severe abdominal pain, vomiting, bloody diarrhea, kidney failure and Hemolytic Uremic Syndrome.

Common Cause: Usually transferred to food by contact with intestines during slaughter.

Apples used for juice where cattle or deer also grazed. Infected employees that do not properly wash their hands after going to the toilet.

Foods Implicated: undercooked beef and other red meats, improperly pasteurized milk, raw finfish,

Prevention: Thoroughly cook ground meats to 165°F, make sure employees properly wash their hands, Use only pasteurized apple cider, fruit juice, and milk products.

Food Allergens

Source: foodborne_illness_ref.pdf, PDF page 11

Type of Illness: An allergic reaction usually involving the skin, mouth, digestive tract, or airways.

Symptoms:

-Skin: hives, rashes, and itching -Mouth: swelling and itching of the lips and tongue -Digestive tract: vomiting and diarrhea -Airways: difficulty breathing, wheezing

Incubation Period: As little as 5 minutes

Common Cause:

Foods containing:

Milk

Egg

Wheat proteins

Peanuts

Tree nuts

Soy

Fish

Shellfish

Prevention: Avoid eating foods containing these ingredients, proper labeling of packaged and prepared foods.

Hepatitis A

Source: foodborne_illness_ref.pdf, PDF page 12

Pathogen: Hepatitis A virus

Type: Viral infection

Oxygen:

Spore Former: No

Incubation Period: 15 to 50 days, average 28-30 days

Duration of Illness: Several weeks, severe cases may last several months

Symptoms: Sudden onset of fever, nausea, vomiting, abdominal pain, fatigue, swelling of liver, and jaundice.

Common Cause: Ingestion of food and water that contain the virus.

Foods Implicated: Raw and lightly cooked shellfish harvested from polluted waters.

Raw vegetables that have been washed with contaminated water. Most Potentially Hazardous Foods that are mishandled by infected food workers.

Prevention: Only use potable water when working with food. Handle foods properly and cook them to proper temperatures. Avoid consumption of raw seafood. Make sure food handlers thoroughly wash their hands.

Listeriosis

Source: foodborne_illness_ref.pdf, PDF page 13

Pathogen: Listeria monocytogenes

Type: Bacterial infection

Oxygen: Facultative anaerobic

Spore Former: No

Incubation Period: 1 day to 3 weeks

Duration of Illness: Indefinite depending on treatment, can be life threatening for pregnant and immuno-compromised.

Symptoms: Flu-like symptoms in healthy adults including nausea, vomiting, headache, fever, chills, and backache. Can be life threatening (septicemia, meningitis, encephalitis, and birth defects) for pregnant woman or immuno-compromised.

Common Cause: Cross contamination or undercooked foods.

Foods Implicated: Raw milk and cheese, raw meats, refrigerated ready-to-eat meats such as hot dogs, raw vegetables, and seafood.

Prevention: Cooked foods thoroughly. Use refrigerated ready-to-eat foods such as cooked turkey breasts, hot dogs, ham, and lunchmeats in a timely manner.

Norovirus

Source: foodborne_illness_ref.pdf, PDF page 14

Pathogen: caliciviruses (previously known as “Norwalk-like viruses”)

Type: Viral infection

Oxygen:

Spore Former: No

Incubation Period: Usually 1 to 2 days, as early as 12 hours

Duration of Illness: 1 to 2 days

Symptoms: Gastroenteritis (“stomach flu”) including nausea, vomiting, diarrhea, abdominal pain, headache, low-grade fever, chills, muscle aches.

Common Cause: Ingestion of food and water that have been contaminated with feces that contain the virus.

Foods Implicated: Raw and lightly cooked shellfish harvested from polluted waters.

Raw salad ingredients. Foods that are handled by infected food workers.

Prevention: Only use potable water when working with food. Handle foods properly and cook them to proper temperatures. Avoid consumption of raw seafood. Make sure food handlers thoroughly wash their hands.

Salmonellosis

Source: foodborne_illness_ref.pdf, PDF page 15

Pathogen: Salmonella spp.

Type: Bacterial infection

Oxygen: Facultative anaerobic

Spore Former: No

Incubation Period: 6 to 72 hours, usually 12 to 36 hours

Duration of Illness: 2 to 3 days

Symptoms: Abdominal pain, headache, nausea, vomiting, fever, diarrhea, dehydration.

Common Cause: Commonly through cross contamination when fecal material is transferred to food through contact with raw foods (especially poultry), contaminated food contact surfaces, or infected food workers.

Foods Implicated: Raw meat and poultry products, shell eggs, sliced fruits and vegetables, milk, dairy products, and other protein foods.

Prevention: Cook foods thoroughly. Clean and sanitize raw food contact surfaces after use. Make sure food workers wash their hands before working with food.

Scombroid Poisoning

Source: foodborne_illness_ref.pdf, PDF page 16

Pathogen: Scombrotoxin

Type: Biochemical intoxication

Oxygen:

Spore Former: No

Incubation Period: A few minutes to 30 minutes after consumption

Duration of Illness: 8 to 12 hours

Symptoms: Dizziness, burning or tingling sensation in mouth, facial rash or hives, peppery taste in mouth, shortness of breath, headache, itchy, teary eyes.

Common Cause: Leaving fish in the Danger Zone.

Foods Implicated: Fish, including tuna, anchovies, blue fish, mackerel, amberjack, abalone, and mahi-mahi. Swiss cheese.

Prevention: Purchase seafood from approved dealers. Keep “fresh” seafood out of the Danger Zone. Do not accept seafood that is suspected of being thawed and refrozen or temperature abused.

Shigellosis

Source: foodborne_illness_ref.pdf, PDF page 17

Pathogen: Shigella

Type: Bacterial infection

Oxygen: Facultative anaerobic

Spore Former: No

Incubation Period: 1 to 7 days, usually 1 to 3 days

Duration of Illness: 4 to 7 days, can be indefinite depending on when treatment is administered.

Symptoms: Diarrhea (sometimes bloody), fever, abdominal cramps, chills, fatigue, and dehydration.

Common Cause: Water that is contaminated by fecal material and food and utensils handled by infected food workers.

Foods Implicated: Common in ready-to-eat salads (i.e., potato, chicken), milk and dairy products, raw vegetables.

Prevention: Cook foods thoroughly, do not allow food workers that have been diagnosed with shigellosis to handle food, make sure food workers wash their hands before working with food, avoid cross contamination, and wash foods with potable water.

Staphylococcus

Source: foodborne_illness_ref.pdf, PDF page 18

Pathogen: Staphylococcus aureus

Type: Bacterial intoxication

Oxygen: Facultative anaerobic

Spore Former: No

Incubation Period: 1 to 6 hours, usually 2 to 4 hours

Duration of Illness: 1 to 2 days

Symptoms: Nausea, vomiting, acute abdominal pains, diarrhea, changes in blood pressure and heart rate.

Common Cause: Contamination from food worker’s hands is the most common way the organism is introduced into foods.

Foods Implicated: Cooked ready-to-eat foods, salads, meat, poultry, custards, high-salt foods, milk and dairy products.

Prevention: Avoid contact of foods with bare hands, Make sure food workers wash their hands and bandage and glove open wounds on their hands. Heat and cool foods properly.

Vibrio spp.

Source: foodborne_illness_ref.pdf, PDF page 19

Pathogen: Vibrio vulnificus, Vibrio parahaemolyticus, Vibrio cholera

Type: Bacterial infection

Oxygen:

Spore Former: No

Incubation Period: 2 hours to 2 days

Duration of Illness: 2 to 3 days, can last longer

Symptoms: Headache, fever, chills, diarrhea, vomiting, skin lesions, abdominal pain, septic shock.

Common Cause: Consumption of raw or lightly cooked seafood.

Foods Implicated: Fish and shellfish, including oysters, clams, mussels, crabs, shrimp, and lobster.

Prevention: Buy seafood, especially shellfish, from approved sources. Cook seafood and shellfish to proper temperatures and avoid eating raw or lightly cooked seafood and shellfish. Avoid cross contamination.

Person’s experiencing any of these symptoms after eating these foods should seek medical attention immediately.